Task 5: Exploratory Data Analysis (EDA)

Titanic Passenger Dataset — Findings Report

Objective: Explore the Titanic passenger dataset using descriptive statistics and visual analysis to identify patterns, trends, missing values and unusual observations.

Dataset overview

Item	Value
Rows / passengers	891
Original columns	12
Survived	342
Did not survive	549
Overall survival rate	38.4%

Data quality observations

Missing values were found in: Age: 177 missing (19.9%); Cabin: 687 missing (77.1%); Embarked: 2 missing (0.2%). Cabin has the largest amount of missing information. Age and Embarked also contain missing values. No duplicate rows were detected.

Descriptive statistics

Median passenger age: 28.0 years. Median fare: 14.45. The age and fare distributions are not perfectly symmetric; fare is strongly right-skewed due to a small number of high fares.

Visual analysis and observations

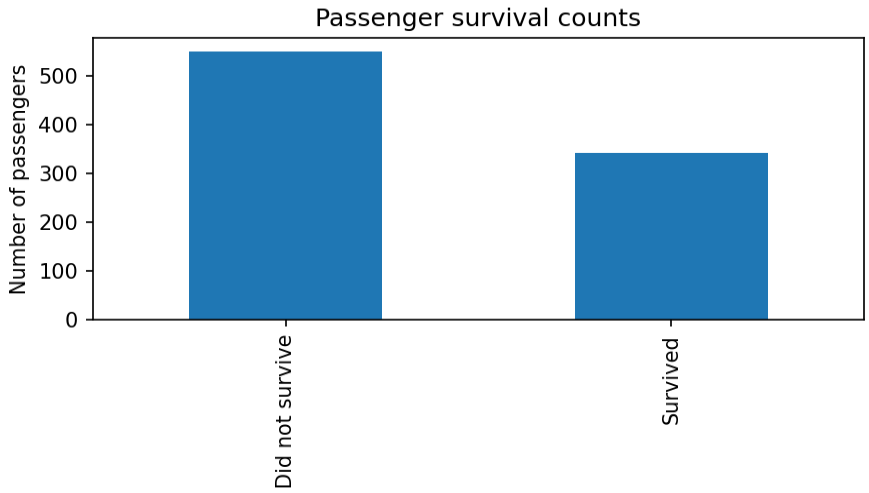


Figure 1. Survival counts. More passengers did not survive than survived.

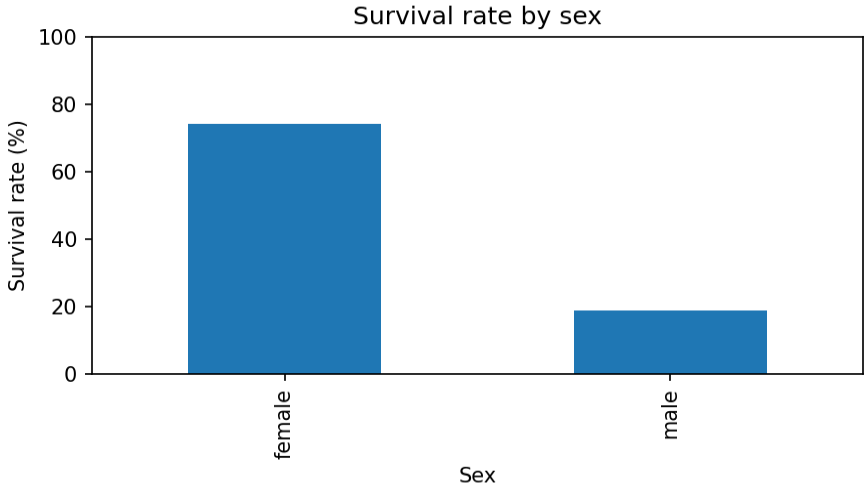


Figure 2. Survival by sex. Female survival rate: 74.2%; male survival rate: 18.9%. This is a large difference in the sample.

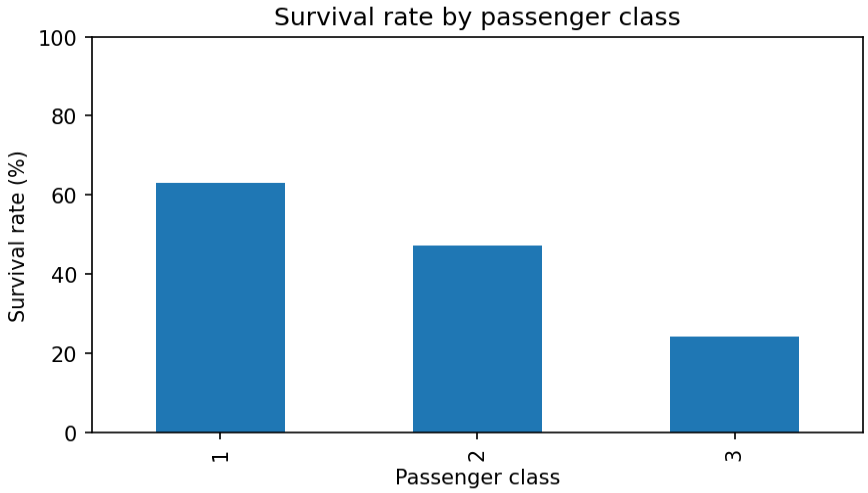


Figure 3. Survival by passenger class. First class has the highest survival rate; third class has the lowest.

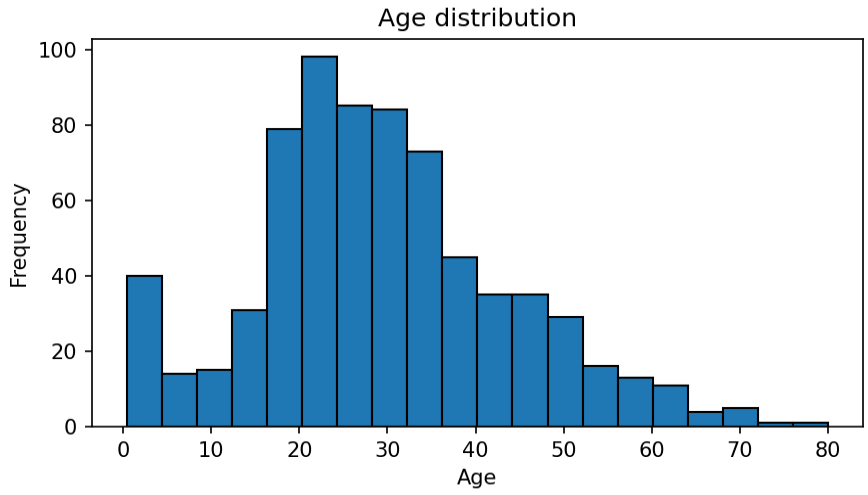


Figure 4. Age distribution. Most recorded ages are concentrated among young and middle-aged passengers.

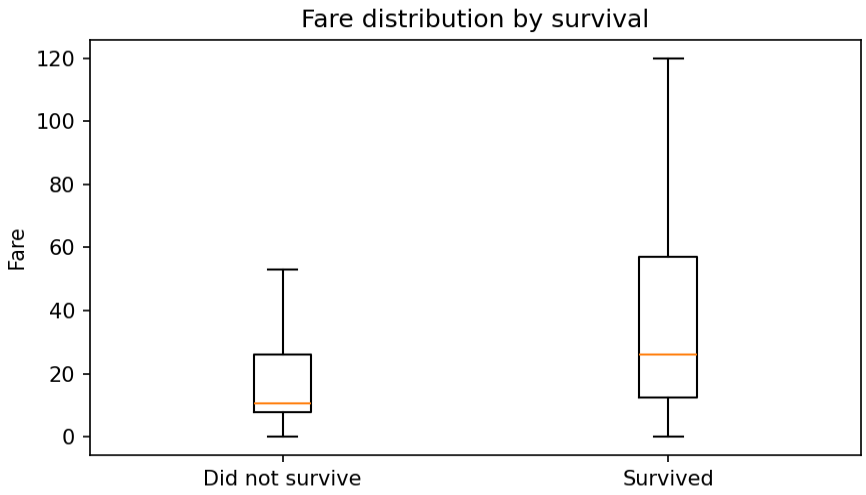


Figure 5. Fare by survival status. Survivors generally have a higher fare distribution, though values overlap.

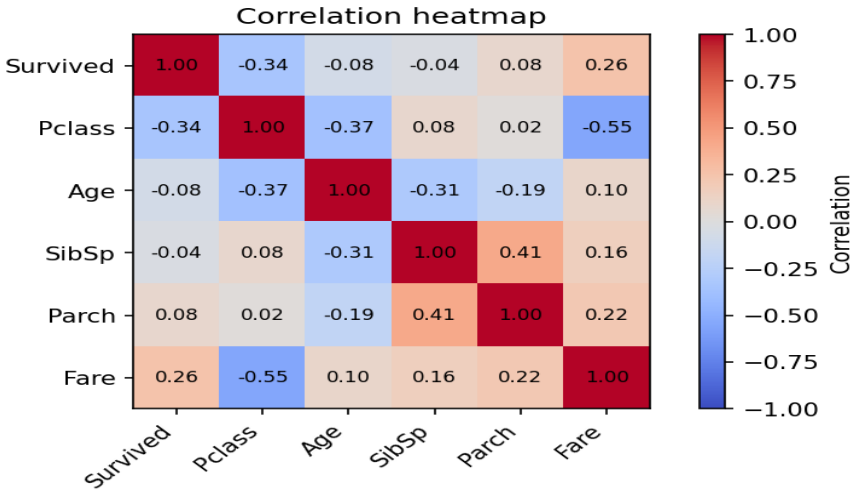


Figure 6. Correlation heatmap. Survival is negatively associated with Pclass and positively associated with Fare.

Group comparison

Group	Survival rate
Sex: female	74.2%
Sex: male	18.9%
Passenger class: 1	63.0%
Passenger class: 2	47.3%
Passenger class: 3	24.2%

Key findings and conclusion

• Overall survival rate was 38.4%, so most passengers in this dataset did not survive.

• Female passengers had a 74.2% survival rate compared with 18.9% for male passengers.

• Passenger class is associated with survival: class 1 had 63.0% survival, class 2 had 47.3%, and class 3 had 24.2%.

• Fare is right-skewed and contains unusually high values; these should be investigated rather than automatically deleted.

• Cabin, Age and Embarked contain missing data. Any later modelling should document the missing-value strategy.

Limitations: This report describes associations in the available sample. Correlation does not prove causation. Passenger class, fare, sex and other variables may be interrelated. The analysis does not claim to explain all causes of survival.

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn and ReportLab. The accompanying Jupyter notebook contains the reproducible analysis code.